

Two dimensional arrays are the most simple form of multidimensional arrays. Here we can imagine that the array elements are structured in both rows and columns. If the rows are the first dimension of the arrays then we can define the columns as the second dimension of the array. Let us look at the example below:

Say we are processing an array that includes three rows and two columns. This is a spreadsheet concerning products. Here, each row is used for each individual product. Among the two columns one is used for the quantity and the other is used for the price of the product. We can treat the elements as columns. Now look at the example below:

```
var row1 = [5, 3.99];    // Quantity 5, Price 3.99
var row2 = [3, 8.99];    // Quantity 3, Price 8.99
var row3 = [2, 60.99];   // Quantity 2, Price 60.99
```

Now let us place the rows into a container array that is used as a container and named as the spreadsheet:

```
var spreadsheet = [row1, row2, row3];
```

The total cost of the order can be searched if we multiply the amount and the cost of each row. We could also add all these together. The elements of a two dimensional Array can be accessed by using two indices where one index is used for the row and the other Index is used for the column. Here the first row's array of two columns can be represented by the expression `spreadsheet[0]`. We can use `spreadsheet[0][1]` in order to access the second column in the first row of the spreadsheet. Look at the following:

```
// Create a variable to store the total cost of
the order
```

```
var total;
```

```
// Now find the cost of the order. For each row,
multiply the columns
```

```
// together, and add that to the total.
```

```
for (var i = 0; i < spreadsheet.length; i++) {
    sum += spreadsheet[i][0] * spreadsheet[i][1];
}
```

```
trace(total);    // Displays: 120.99
```

Multidimensional arrays can behave just like other normal arrays. We may happily use all of the array methods to operate on the data that are stored in a multidimensional array.